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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 16: Global practices in EIA process

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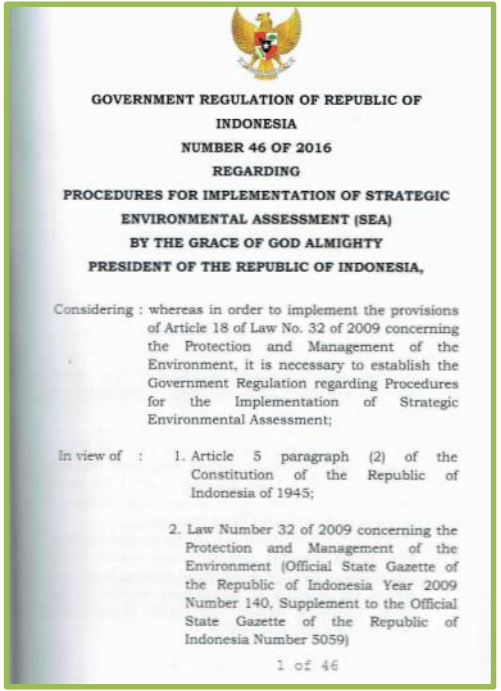


Contents

- Approaches for different stages of EIA process in various countries
  - Requirement of Strategic Environment Assessment
  - Selection of parameters for Screening process
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  - Public participation process
  - Cultural impact assessments
- Comparative overview of EIA process in the developed & developing countries & India
- Gaps/Loopholes in EIA process
- Conclusion

Requirement of Strategic Environment Assessment (SEA) in different countries

- Legal provisions have been integrated within EIA legislation
  - Ex.- Kenya (Africa) and Tanzania (Africa)
- Separate laws or regulations for SEA
  - Ex.- Indonesia (South-east Asia) and Canada (North America)
- Legal requirements have been integrated within other Sectoral legislations (Transport, Energy etc.)
  - Ex.- New Zealand



GOVERNMENT REGULATION OF REPUBLIC OF INDONESIA
NUMBER 46 OF 2016
REGARDING
PROCEDURES FOR IMPLEMENTATION OF STRATEGIC ENVIRONMENTAL ASSESSMENT (SEA)
BY THE GRACE OF GOD ALMIGHTY
PRESIDENT OF THE REPUBLIC OF INDONESIA,

Considering : whereas in order to implement the provisions of Article 18 of Law No. 32 of 2009 concerning the Protection and Management of the Environment, it is necessary to establish the Government Regulation regarding Procedures for the Implementation of Strategic Environmental Assessment;

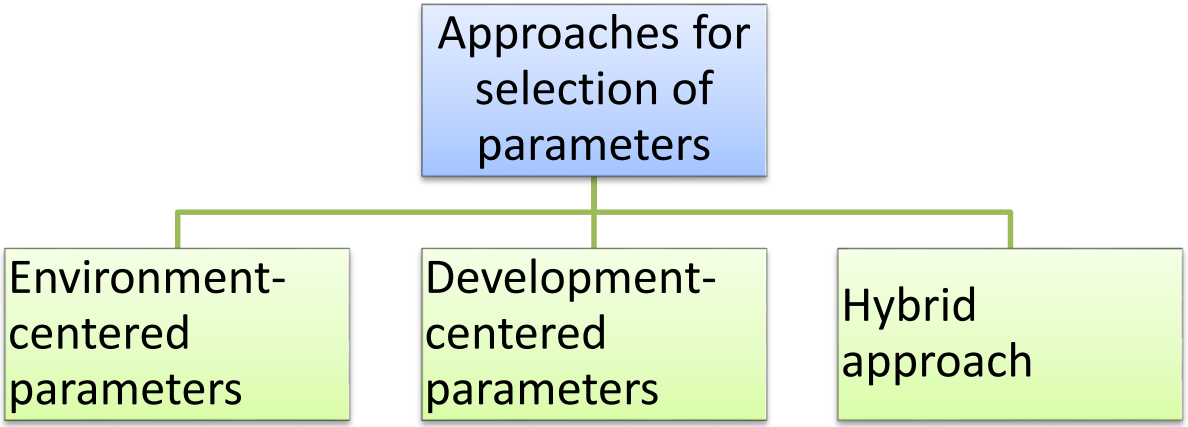
In view of : 1. Article 5 paragraph (2) of the Constitution of the Republic of Indonesia of 1945;

2. Law Number 32 of 2009 concerning the Protection and Management of the Environment (Official State Gazette of the Republic of Indonesia Year 2009 Number 140, Supplement to the Official State Gazette of the Republic of Indonesia Number 5059)

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Image: SEA regulations in Indonesia

Approaches of selecting parameters for EIA



- **Environment-centered:** Listing of factors/criteria/parameters for assessing the likely Impact
- **Development-centered:** Listing development activities
- **Hybrid:** Between two approaches

Environmental impact centered parameters for EIA screening

S.No.	Information/Checklist confirmation	Yes/No	Details thereof (with approximate quantities/rates, wherever possible) with source of information data
5.1	Emissions from combustion of fossil fuels from stationary or mobile sources		
5.2	Emissions from production processes		
5.3	Emissions from materials handling including storage or transport		
5.4	Emissions from construction activities including plant and equipment		
5.5	Dust or odours from handling of materials including construction materials, sewage and waste		

Sample Check list for project screening in India

Would you consider these information asked as Environmental impact centered? (Yes, these are..)

For screening process, India follows hybrid approach

- Information asked in Checklist contains both development/ activity and environmental centered parameters

Development/activity centered parameters for EIA screening

S.No.	Information/Checklist confirmation	Yes/No	Details thereof (with approximate quantities /rates, wherever possible) with source of information data
1.1	Permanent or temporary change in land use, land cover or topography including increase in intensity of land use (with respect to local land use plan)		
1.2	Clearance of existing land, vegetation and buildings?		
1.3	Creation of new land uses?		
1.4	Pre-construction investigations e.g. bore houses, soil testing?		
1.5	Construction works?		

Sample Check list for project screening in India

Would you consider these information asked in the Checklist as development/activity centered? (Yes, these are..)

Parameters for EIA Screening: European Union (EU)

Sample Check list for project screening in EU

- Is this check-list development centered or environment centered?
  - (This is an example of hybrid approach..)

1. Will construction, operation, decommissioning or demolition works of the Project involve actions that will cause physical changes in the locality (topography, land use, changes in waterbodies, etc.)?
2. Will construction or the operation of the Project use natural resources such as land, water, materials or energy, especially any resources which are non-renewable or are in short supply?
3. Will the Project involve the use, storage, transport, handling or production of substances or materials which could be harmful to human health, to the environment or raise concerns about actual or perceived risks to human health?
4. Will the Project produce solid wastes during construction or operation or decommissioning?
5. Will the Project release pollutants or any hazardous, toxic or noxious substances to air or lead to exceeding Ambient Air Quality standards in Directives 2008/50/EC and 2004/107/EC)?

Screening Process for EIA in European Union

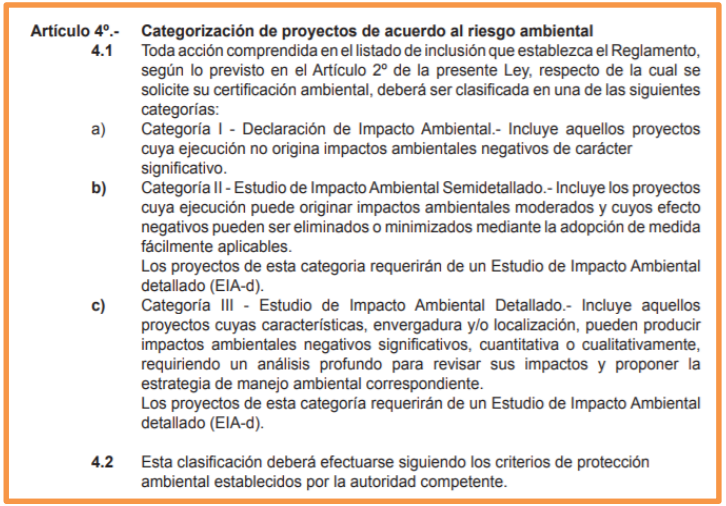
- In the European Union, member countries are required to classify projects in two categories (I & II)
- Determination based on a case-by-case examination or the setting of thresholds criteria or a hybrid of both by countries separately.



Screening Process for EIA in Peru

Screening through three project categories:

- Category-I projects: Only an Environmental Impact Statement (EIS) needs to be prepared, (expected low environmental impacts)
- Category-II projects: Need to undergo a Semi-detailed Environmental Impact Study (EIA-SD) (expected moderate impacts from projects)
- Category- III projects: Need to undergo a Detailed Environmental Impact Study (EIA-D) (expected significant environmental impacts)



Artículo 4º.- Categorización de proyectos de acuerdo al riesgo ambiental

4.1 Toda acción comprendida en el listado de inclusión que establezca el Reglamento, según lo previsto en el Artículo 2º de la presente Ley, respecto de la cual se solicite su certificación ambiental, deberá ser clasificada en una de las siguientes categorías:

a) Categoría I - Declaración de Impacto Ambiental.- Incluye aquellos proyectos cuya ejecución no origina impactos ambientales negativos de carácter significativo.

b) Categoría II - Estudio de Impacto Ambiental Semidetallado.- Incluye los proyectos cuya ejecución puede originar impactos ambientales moderados y cuyos efecto negativos pueden ser eliminados o minimizados mediante la adopción de medida fácilmente aplicables.
Los proyectos de esta categoria requerirán de un Estudio de Impacto Ambiental detallado (EIA-d).

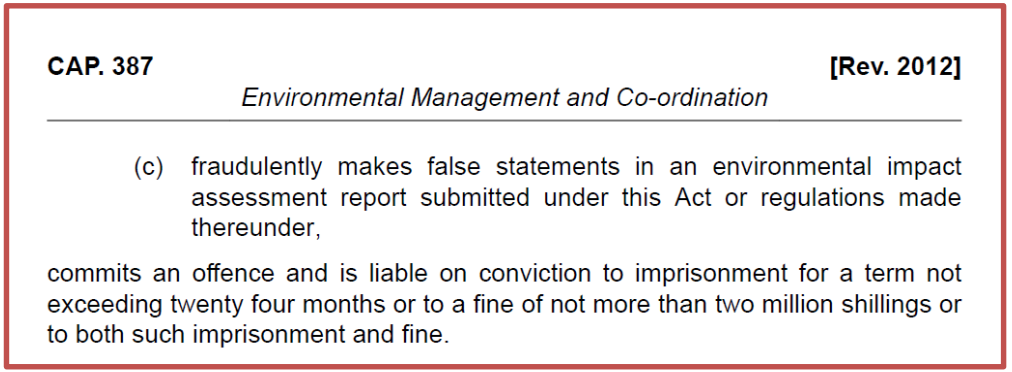
c) Categoría III - Estudio de Impacto Ambiental Detallado.- Incluye aquellos proyectos cuyas características, envergadura y/o localización, pueden producir impactos ambientales negativos significativos, cuantitativa o cualitativamente, requiriendo un análisis profundo para revisar sus impactos y proponer la estrategia de manejo ambiental correspondiente.
Los proyectos de esta categoría requerirán de un Estudio de Impacto Ambiental detallado (EIA-d).

4.2 Esta clasificación deberá efectuarse siguiendo los criterios de protección ambiental establecidos por la autoridad competente.

Image: Categorization of projects, Article 4, Law 27446

Special feature of EIA screening process in Kenya

- The Kenyan EIA legislation is one of the few that requires the involvement of an environmental practitioner at screening stage.
- Knowingly providing false or misleading information is considered an offence and leads to the punishment.



CAP. 387 [Rev. 2012]

Environmental Management and Co-ordination

(c) fraudulently makes false statements in an environmental impact assessment report submitted under this Act or regulations made thereunder,

commits an offence and is liable on conviction to imprisonment for a term not exceeding twenty four months or to a fine of not more than two million shillings or to both such imprisonment and fine.

Image: Article 138, Section C, EIA legislation of Kenya

Consideration of cumulative impacts during scoping stage

Impacts of project are considered in regional perspective rather than local or independent, called cumulative impact.

Ex.- China

- No mention of cumulative impact assessment for EIA

Ex.- Canada

- Study of cumulative effects of physical activities in a region and their consideration in environmental assessment

(ii) in a province other than the one in which the act or thing is done or where the physical activity, the designated project or the project is being carried out, or

(iii) outside Canada; and

Image: Consideration of impact (Sub section ii, iii, section b, article 5, Canadian EIA legislation 2012)

Examples of cumulative impacts

In Megacity Delhi,

- Vehicle and industrial sources are the main contributors of pollution.

- But in winter season, sudden rise in pollution is observed
  - Experts believes that stubble burning in the neighboring states adds a large chunk of pollutants to Delhi's environment.



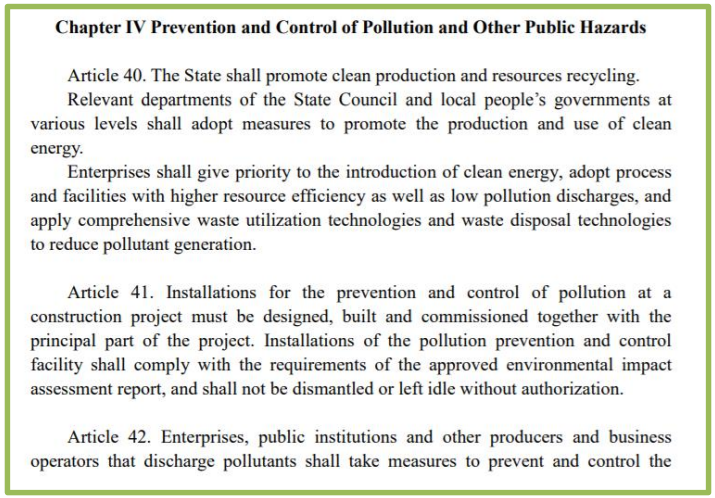
Image source: theprint.in

Do you think, to lower this pollution, Management should be in a macro (Regional) or micro level (state boundaries)?
(Cumulative impacts needs to be addressed at regional levels..)

Mitigation Measures for individual impacts

Example: China

- Mitigation measures for each likely adverse environmental impact should considered during scoping process.
- Installations for the prevention and control of pollution at a construction project must be designed, built and commissioned.



Chapter IV Prevention and Control of Pollution and Other Public Hazards

Article 40. The State shall promote clean production and resources recycling.

Relevant departments of the State Council and local people's governments at various levels shall adopt measures to promote the production and use of clean energy.

Enterprises shall give priority to the introduction of clean energy, adopt process and facilities with higher resource efficiency as well as low pollution discharges, and apply comprehensive waste utilization technologies and waste disposal technologies to reduce pollutant generation.

Article 41. Installations for the prevention and control of pollution at a construction project must be designed, built and commissioned together with the principal part of the project. Installations of the pollution prevention and control facility shall comply with the requirements of the approved environmental impact assessment report, and shall not be dismantled or left idle without authorization.

Article 42. Enterprises, public institutions and other producers and business operators that discharge pollutants shall take measures to prevent and control the

Image: Article 40,41,42; Environment Protection Act, 2014

Example of mitigation measure for individual impacts

- Surrounding the noise making facility/machine/road with noise barriers
- Treatment of sludge of factories before dumping it into rivers.
- Using efficient machines in a manufacturing unit (less fuel consumption & pollution)



Image: Noise barrier along the road

Image source: perforated-pipe.com,

Offset for adverse environment impacts

Example: Brazil

- Projects subject to environmental licensing must offset their impacts on protected areas.

Example: Mexico

- Sustainable Forest Development Act of 2003 requires offsets for impacts that result in land-use change to forested areas.

- Recently enacted Environmental Liability Act (2013) also requires offsets, but only when impacts are not predicted or approved in the EIA and are an environmental offence.

Example of offset for impacts: Antamina mines

- Polylepis (a shrub) habitat is indirectly impacted by the Antamina mine
- The EIA concludes that overall impacts to flora and fauna are limited. The key concern identified in the EIA was the potential for destruction of the endangered shrub species (Polylepis)
- Conservation programme was carried out at large scale



Image: Antamina Mines in Peru
Source: bhp.com

Responsibility of preparing EIA reports

- Government agency

Ex.- U.S.A.

- Government licensed or registered consultants or agencies

Ex.- Vanuatu, Cameroon, Peru, China, Mongolia, Oman, Lebanon

- Individual with certification

Ex.- Indonesia

In India, EIA reports are prepared by agencies registered under QCI (Quality Council of India) or notified by Ministry of Environment, Forest & Climate Change (MoEF&CC)

Public Participation in Canada

- Public is invited to submit comments for a period of 20 days after project is subject to a screening & similarly at other stages.
- Any comments received must be considered by the Canadian Environmental Assessment Agency.

In Canada, public participation is a multi-stage process.

- It is starts from screening stage up to monitoring stage which is against general practice in most of the other countries.

Definition of Public 1/2

Example: China

- 2014 Environmental Protection act: make public the full text of the EIA report.

- Opportunity to participate in hearings or to make oral or written submissions to the Pollution Control Boards is restricted to residents, environmental groups and others located at the project site and likely to be affected.

Article 57. Citizens, legal persons and other organizations shall be entitled to report and complain environmental pollution and ecological damage activities of any units and individuals to competent environmental protection administrations or other departments with environmental supervision responsibilities.

In the event the local people's government and its environmental protection administrations or any other relevant departments fail to fulfill their responsibilities in accordance with the law, any citizen, legal person or other organizations have the right to report it to the competent higher level governments or the supervisory department according to law.

The authorities receiving the report shall keep confidential the relevant information of the informant, and protect the legitimate rights and interests of the informant.

Article 58. For activities that cause environmental pollution, ecological damage and public interest harm, social organizations that meet the following conditions may file litigation to the people's courts:

(1) Have their registration at the civil affair departments of people's governments at or above municipal level with sub-districts in accordance with the law;

(2) Specialize in environmental protection public interest activities for five

Image: Article 57, 58; Environment Protection Act, 2014

Definition of Public 2/2



Example: Denmark

- One or more natural or legal persons (companies, etc.) directly or indirectly affected or likely to be affected

- Associations and organizations that have the protection of the landscape, cultural heritage, natural or environmental interests as objective

Example of public participation: Light Rail Project, Baltimore, U.S.A.

- 52 Pre-scoping “Listening Tours”
- Initial scoping meetings/written comment period
- Published reports on website
- Designation of community liaisons
- Four Public hearings
- Citizens' Advisory Council
- Speaker's Bureau Meetings/Neighborhood Events/Community Meetings
- Study Area Resource Hubs and designated liaisons for specific communities

Source: epa.gov

EIA Review process in the European Union (EU)

- EU guideline suggests to review the EIS (Environment Impact Statement) report before any decision making.

- Review should be done either by govt. agencies or professional asked to do so on behalf of government.

A3 Implementation of Review in the EU

In some Member States review of the adequacy of EIS before they are used for decision-making is a mandatory stage in the EIA procedure. In these cases the review may be undertaken by the competent authority or by an independent organisation on behalf of the competent authority. Where the EIS is considered to be inadequate, the developer will be asked to provide additional information and the development consent decision process will not start until this information has been provided. There will usually be a procedure for appeal against requests for further information.

In other Member States there is no formal stage of review in the EIA procedure but competent authorities will usually undertake some sort of review before starting the decision-making process, to ensure that the requirements of the legislation have been met. They will then usually have the power to ask for further information from developers before the decision-making process starts, if they consider the EIS to be inadequate. Review may also be undertaken informally by the developer prior to submitting the EIS to the competent authority or by consultees after it is submitted, to check that the information is adequate.

Various professional bodies, for example the Institute of Environmental Management and Assessment in the United Kingdom (http://www.iem.org.uk) and EIA research institutions (see http://europa.eu.int/comm/environment/eia/contacts2.htm for a list of EIA Centres) will provide EIS review as a service to competent authorities, developers and other parties if required.

An overview of Member State requirements for review as they were in August 1999 can be found in a report at *http://europa.eu.int/comm/environment/eia/eia-studies-and-reports/study1.htm.*

Image: Section A3, Eu guidelines for review

EIA Review process in Cameroon

- After the Environmental and Social Impact Assessment (ESIA), report submits to the Ministry of Environment, Nature Protection and Sustainable Development
- Ministry can undertake on-site visits to qualitatively verify the information included in the assessments and to obtain feedback by the concerned public
- The outcomes are transmitted in a report to the Inter-ministerial Committee.



Global best features for follow up/monitoring as suggested by the UNEP (United Nations Environment Programme)

- Sufficient information available at regular intervals
- Manage adverse impacts based on monitoring data (if any)
- Ensure indigenous community participation in environmental management
- Expend the powers of the competent enforcement agencies
- Create regional monitoring networks

Special assessment of cultural impacts in EIA process:

Example of Canada

(c) with respect to aboriginal peoples, an effect occurring in Canada of any change that may be caused to the environment on

(i) health and socio-economic conditions,

(ii) physical and cultural heritage,

(iii) the current use of lands and resources for traditional purposes, or

(iv) any structure, site or thing that is of historical, archaeological, paleontological or architectural significance.

Image: Section C, Article 5, Canadian EIA legislation, 2012

Example of Laurentia Project in Canada

Laurentia Project: Port of Quebec Deep-Water Wharf – Beauport

EIA report concluded: Significant adverse environmental effects to fish and fish habitat, air quality and human health, socio-economic conditions, and the current use of lands and resources for traditional purposed by Indigenous peoples.

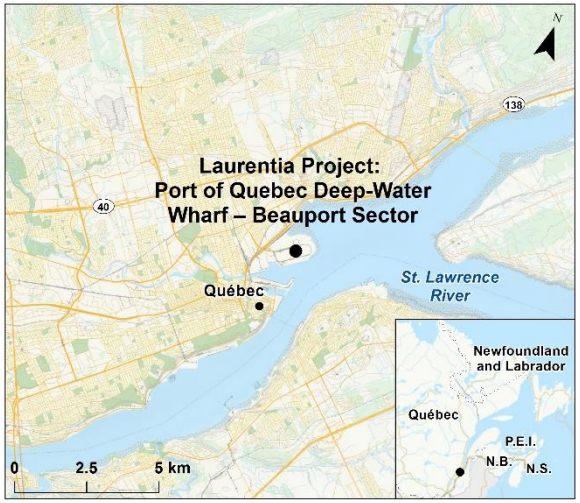


Image: Project site location for project

- Ultimately, this project was not commissioned.

Special assessment of cultural impacts in EIA process:

Example of New Zealand

For aboriginal peoples (indigenous tribal people),
environmental effects that are to be considered in EIAs are:

- Health and socio-economic conditions;
- Physical and cultural heritage;
- Current use of lands and resources for traditional purposes
- Structure, site or objective that is of historical, archaeological, paleontological or architectural significance
- Impact on community knowledge and aboriginal traditional knowledge.

Formal & legal basis: Comparative Overview of EIA process

Developed Countries	Developing Countries	India
• Well-framed EIA legislation in place. For instance, in Canada, Canadian Environmental Assessment Act regulates EIA, while EU countries are guided by Directive on EIA (1985)	• Lack of formal EIA legislation in many developing countries. For instance, EIA is not mandatory in many African countries	• Formal legislation for EIA. It has been enacted by making an amendment in the Environment Protection Act 1986.

Source: Industry & Environment Unit, Centre for Science & Environment

Public Participation: Comparative Overview of EIA process

Developed Countries	Developing Countries	India
• Active involvement of all participants including competent authority, government agencies and affected people at early stages of the EIA. This makes the process more robust and gives a fair idea of issues, which need to be addressed in the initial phase of EIA.	• Limited involvement of public and government agencies in the initial phases. This often results in poor representation of the issues and impacts in the report, adversely affecting the quality of the report.	• Limited involvement of public and government agencies in the initial phases.

Impact type & Alternatives: Comparative Overview of EIA process

Developed Countries	Developing Countries	India
• Integrated approach to EIA followed. All aspects including social and health considered.	• Mainly environmental aspects considered. Poor on social or health aspects.	• No provision in place to cover landscape and visual impacts in the Indian EIA regulations
• Proper consideration of alternatives in EIA	• The consideration of alternatives in developing countries is absent.	• Same as developing countries

Language & technical approach: Comparative Overview of EIA process

Developed Countries	Developing Countries	India
• Most reports in local language	• Most reports in one primary language and not in the local language.	• Most reports in English and not in the local language. In some case, executive summary is translated in local language.
• A multi-disciplinary approach. Involvement of expert with expertise in different areas.	• Lack of trained EIA professionals often leads to the preparation of inadequate and irrelevant EIA reports in developing countries	• Preparation of EIA is done by consultants. Therefore, the selection criterion for the organization is fees/cost rather than the expertise of EIA team.

Expertise level: Comparative overview of EIA process

Developed Countries	Developing Countries	India
• Many individuals with the capability to design, conduct, review and evaluate EIAs from countries of the North. The major portion of teaching about environmental assessment also takes place in industrial countries.	• The expertise in EIA is slowly developing. In most cases, students from the developing countries go to the developed countries to gain knowledge of the subject.	• Expertise in this area is developing.

Gaps/Loopholes in EIA process: Project splitting or Salami Slicing approach

To by-pass the EIA process all together:

- Separate the project into small parts which do not cross threshold level individually but as whole may come into high impact projects
- This is very common practice in various countries where projects are splitted into many small projects

Example of Project splitting in Spain

- Failing to carry out an assessment of the effects on the environment (through project splitting) of the 'project for a Valencia-Tarragona railway line, Las Palmas-Oropesa section' which forms part of the project known as the 'Mediterranean corridor,
- Spain was forced to carry out EIA

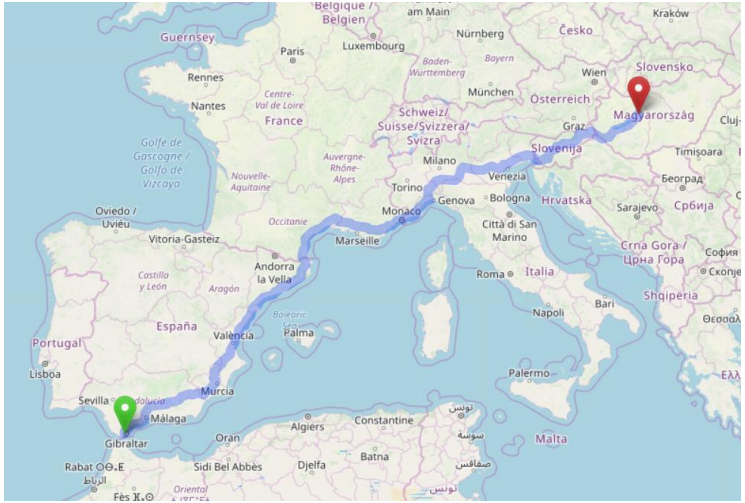


Image: Mediterranean Corridor Rail line map

Example of Project splitting in U.K.

- Urban development projects at White City and at Crystal Palace, project was divided into small activities.
- The United Kingdom of Great Britain and Northern Ireland has failed to fulfil its obligations to carry out the EIA process.
- Project was stayed by European court



Image: Project site of White city project

Measures to avoid project splitting: European Countries

Country	Regulations
Portugal	• Apply EIA below the thresholds if necessary
Sweden	• Assess the scope of the project in the screening phase
Denmark, Finland, Spain	• Use case-by-case screening
Germany, U.K.	• Regulation for requirements to consider a project as independent
Austria	• Consider all the associated developments part of a project

Source: EC (2003), IMPEL (2012), Salamanca A. (2016)

Gaps/Loopholes in EIA process: Examples of exemption from EIA

Example of Nigeria

- Projects are exempted if it is in the interest of public health or safety

Example of Indonesia

- An activity can be exempted from the obligation to conduct an EIA if the location of the activity is within an area that has had an EIA already
- Location of the activity is within a district/ city that has already adopted a detailed spatial plan and/or a district/city strategic area spatial plan.

Conclusion

- The approaches for for different stages of EIA vary from country to country.
- Best practices of EIA from countries worldwide can be adopted by other countries to minimize the significant impacts on the environment.
- More constructive public participation is required to be adopted during the EIA process to ensure transparency and increase acceptability of the projects.
- EIA process need to be implemented in spirit, not just words.

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Thank You